

Performance Evaluation of Battery-Operated Three-Wheeler Chassis by Finite Element Analysis

Harshvardhan Kumbhar^(1,a), Om Kotagi^(1,b), Shakir Gul^(1,c), Aditya Pisale^(1,d),
Gautam Shahapurkar^(1,e), S. B. Angadi^(1,f)

Author Affiliations

¹, Department of Mechanical Engineering, KLE Technological University, Dr. M. S. Sheshgiri Campus, Belagavi, India

^{a)} kumbharharshvardhan6@gmail.com

^{b)} omkotagi10@gmail.com

^{c)} wanishakir217@gmail.com

^{d)} adityapisale549@gmail.com

^{e)} gautamshahapurkar@gmail.com

^{f)} sbangadi2003@gmail.com

Abstract. Battery-operated three-wheelers, also known as electric trike, are acquiring fame as a process of convey for both trade and traveler intention in India. The chassis indicates to the base pillar established at the base of the vehicle. This frame, typically constructed from steel, acts as the center layout, holding all crucial elements of the vehicle. The chassis frame must have good solidity to tolerate various forces, such as non-dynamic forces, inertia effects, and forces emerging from acceleration and decelerating. Moreover, it should provide applicable bending and twist resistance to verify stability and efficient handling. This study focuses on the non-dynamic analysis of the chassis frame used in 3-wheeler EV. CATIA software is used to make the 3D diagrams, and its non-dynamic implementation was analyzed through finite element analysis (FEA) in ANSYS software. The analysis checks key factors such as the deformation, equivalent strain, equivalent stress and the maximum principal stress of the chassis developed during static circumstances. Material used to implement the design are Aluminum 6061-T6, Magnesium Alloy AZ31B, Low Alloy Steel 4140, Grey Cast Iron ASTM A48 and Carbon Steel 1020. The optimized weight in the optimized chassis is 944N. It has been observed that Low Alloy Steel is the best material overall due to its combination of low deformation, low strain and high rigidity.

Keywords: Chassis frame, static analysis, finite element analysis, deformation, equivalent stress, equivalent strain and maximum principal stress.

1. INTRODUCTION

The number of SI and CI vehicles has raised greatly in both domestic and business transport across cities and towns. This increase in vehicle application has led to severe air and noise pollution in many urban and rural areas. Such pollution harmfully impacts human health and the environment, assisting to issues like global warming and rising sea levels. To address this, battery-operated three-wheelers have developed as a positive solution for non-polluting, noise-free transportation in both urban and rural regions of India. These electric vehicles are set to become a possible option to traditional SI/CI-operated vehicles.

The motorized chassis plays a critical role in any traveler or business-related vehicles, fulfilling as the foundation that supports various elements, including delay, slowing, and directing systems. Typically forming the bottom of the chassis is constructed with joined by a series of side and cross members. Additionally, vertical members supporting the directing system are critical components of the chassis structure. A chassis frame must control sufficient strength to withstand static load conditions without too much deflection or deformation. At the same time, it should be as lightweight as possible to verify efficiency, without compromising its strength or rigidity.

Several researchers have conducted extensive studies on the design and analysis of three-wheeler battery-operated chassis, exploring various material combinations and structural configurations. The EV 3-wheeler chassis was analyzed manually as a simply supported beam under UDL and validated using ANSYS FEA software. The static condition deformation was 0.764 mm, and the highest stress was 137.33 MPa, as per FEA, compared to theoretical values of 2.1 mm deflection and 94.16 MPa stress. The chassis was designed in CATIA and analyzed using ANSYS, with results confirming safety within material strength limits. The material they have used is IS2062 grade steel. [1] The dynamic properties of a three-wheeled electric vehicle with a focus on driving safety. They analyze stability during various driving conditions, including acceleration, braking, and cornering, to identify potential tipping risks. The study evaluates the influence of the vehicle's unique design and chassis configuration on stability and control. They use simulation tools and experimental data to validate their findings. Recommendations are provided to optimize the chassis design for improved safety and performance.[2] They focuses on the design and development of a three-wheel multi-purpose electric vehicle,

incorporating finite element analysis (FEA) for structural evaluation. They design the vehicle with multi-purpose functionality to meet diverse usage needs while optimizing performance and safety. The study utilizes FEA to analyze and validate the structural integrity and durability of key components under different loading conditions. Prototypes are developed to test and refine the design based on simulation results. The research aims to provide a cost-effective and reliable electric vehicle solution for varied applications. The material generally they are testing in structural steel.[3] They focuses on designing and fabricating a prototype battery electric three-wheeled vehicle for urban mobility. They optimize key components such as the electric motor, battery system, and lightweight chassis to enhance energy efficiency and performance. The prototype is tested for stability, range, and safety under real-world conditions. Emphasis is placed on cost-effectiveness, environmental sustainability, and practicality for daily use in cities. The material they used is aluminum 6061-T6 and have analyzed through FEM. [4] They design of a battery module case for electric vehicles using topographical optimization techniques. The study aims to reduce the weight and material usage while maintaining the structural integrity and safety of the battery case. They employ advanced simulation methods to identify the most efficient design geometry that improves performance and cost-effectiveness. The optimization process ensures better load distribution and energy efficiency. The research provides insights into enhancing the overall efficiency and sustainability of electric vehicle battery system. The material that they tested is aluminum alloys.[5] They focus on designing the chassis and battery pack for an electric skateboard vehicle. They perform structural analysis to optimize the chassis for strength, stability, and weight reduction. The battery pack design is optimized for energy efficiency, safety, and compactness. The research incorporates simulation tools to evaluate performance under different load and stress conditions. The study aims to provide an efficient, durable, and lightweight design for electric skateboard applications. The material they used is aluminum 6061-T6 and uses ANSYS software to analysis different structural performances under various conditions. [6] They provide a comprehensive analysis of the key components involved in the design, including the chassis, electric motor, battery systems, and overall vehicle architecture. The review highlights different design strategies, performance considerations, and challenges faced in developing efficient and cost-effective three-wheeled electric vehicles. The authors also assess the environmental impact, safety, and sustainability of such vehicles. The paper aims to provide insights into the latest developments and trends in the design and optimization of three-wheeled EVs for urban and commercial applications. They have done only the literature survey upon the chassis. [7] They focused on designing and optimizing the frame of a three-wheeled electric vehicle (EV). They explore various structural materials to enhance the strength, durability, and weight efficiency of the frame. The study uses optimization techniques to reduce the frame's weight while maintaining its structural integrity and safety. They use AlSiMg alloy or similar material to keep the vehicle lightweight and easy to maneuver. This alloy also provides sufficient strength for everyday use.[8] A brushless DC (BLDC) motor is selected for three-wheeler electric vehicles after evaluating various motor options. The operating characteristics—power, torque, and speed—of the BLDC motor are determined using a longitudinal vehicle dynamics (LVD) model based on the existing Indian driving cycle (IDC). MATLAB simulations are used to compute the motor's power, torque, and speed requirements for both lead-acid and lithium-ion battery systems. The study calculates peak and RMS torque and power values, as well as the motor's operating speed. Finally, the results are compared with literature models to optimize the e-propulsion system's performance. This research is based on the pure electrical based. [9] The chassis design is optimized for better load-bearing qualities and reduced weight. Finite element (FE) simulations are conducted to analyze the chassis' equivalent voltage and deformation under different loading conditions. The results show significant stress concentrations at the chassis edges and center, indicating potential failure under extreme loads. Maximum deformation of 12.047 mm occurs at the chassis' midsection, while the ends show minimal deformation. The study emphasizes the importance of reducing weight for cost-effective production while enhancing the chassis' structural response. The material they have used is structural steel and have done analysis on ANSYS software.[10] The study shows that the EV is powered by both human and electric motor. The vehicle is designed to reduce reliance on fossil fuels while offering an eco-friendly transport solution. SolidWorks and Ansys were used for simulating the vehicle's components, including stress analysis of the frame under static and dynamic conditions. The frame was optimized for strength while minimizing weight using AISI 1020 carbon steel pipes. The vehicle features a PMDC motor and was tested for performance, proving to be an efficient alternative to petrol or diesel vehicles. [11] This paper focuses mainly on the comparison between the maximum stress, deformation and strain of loop type bicycle frame. By varying the centre distance there was no impact on deformation and strain. However, the loop-type frame showed greater deformation compared to a triangle frame due to its less rigid structure. The study also found that intersecting the centrelines of the top and down tubes with the head tube improved rigidity and reduced deformation. Additionally, a magnesium alloy frame was found to have higher displacement and strain energy than an aluminium alloy frame, though the MTB frame using AZ61/Al2O3 MMCs was lighter and structurally rigid. [12] The static structural analysis of a chassis was conducted using ANSYS software. The study compared the performance of different materials, including AISI 4130 Steel, ASTM A710 Steel, grey cast iron, and Al 6063. The results showed that Steel alloy (AISI 4130 and ASTM A710) exhibited the least deformation (1.49 mm) and strain (0.0021). Based on the analysis, it was concluded that Steel alloy are the most suitable materials for automobile chassis manufacturing due to their superior strength and performance. [13] It examines the suitability of materials such as steel, aluminum, and carbon fiber composites for chassis construction. It evaluates these materials based on criteria like strength, weight, stiffness, durability, cost, and sustainability using theoretical analysis, simulations, and practical testing. The study emphasizes the trade-offs between lightweighting and performance, offering insights into material selection for optimizing efficiency and safety in three-wheeled electric vehicles. It highlights the importance of balancing cost-

effectiveness with sustainability goals. The findings guide manufacturers and engineers in developing eco-friendly and efficient electric vehicle designs. From the above literature review, it has been observed that there has been no work carried out based on optimization. [14] The paper Modal Analysis of Three-Wheeler Auto Chassis Using ANSYS investigates the dynamic behavior of a three-wheeler chassis by determining natural frequencies and mode shapes, such as roll, pitch, and tramp modes. It evaluates performance under payload and no-payload conditions to avoid resonance and ensure ride comfort and safety. Using ANSYS simulations, the study identifies critical modes affecting structural stability. The findings offer valuable insights for optimizing chassis design. This research contributes to improving the dynamic performance and safety of three-wheeled vehicles. [15] The paper focuses on designing and analyzing the frame of an electric three-wheeler using ANSYS Workbench. It evaluates the frame's structural integrity by studying stress, strain, and deformation under various loading conditions. The vehicle features a two-front, one-rear wheel configuration and utilizes an electric motor with rechargeable batteries. The study demonstrates the frame's ability to withstand operational forces, highlighting its suitability for electric vehicle applications. It also encourages further research into optimizing vehicle frames using FEA tools. [16] The paper Design, Development and Analysis of a Three-Wheeler Vehicle Chassis explores the development of lightweight chassis designs for energy-efficient vehicles in a tadpole configuration. It evaluates spaceframe, semi-monocoque, and monocoque chassis constructions using finite element analysis (FEA) in ANSYS software, focusing on torsional and bending stiffness. The study compares these designs in terms of performance, fuel economy, manufacturing feasibility, and cost. The research aims to optimize chassis design for improved efficiency and lightweight construction. [17] The paper Design and Development of Chassis for Front-Wheel Drive Electric Tadpole Vehicle focuses on designing a tubular space frame roll cage chassis for a three-wheeled tadpole vehicle. It includes material selection, static and dynamic load calculations, impact analysis, and the development of a CAD model using SolidWorks. Five chassis models were designed and analyzed in ANSYS, with the best one selected based on performance and safety criteria. The tadpole configuration, with two front wheels and one rear wheel, was chosen for its superior dynamic stability, aerodynamics, and braking performance compared to the delta configuration. The study aims to optimize the chassis for efficiency, safety, and manufacturability. [18] From the above literature review, it has been observed that there has been no work carried out based on optimization.

2. THEROTICAL CALCULATION OF EXISTING CHASSIS:

The chassis frame is the foundation of a vehicle, determining its shape, size, and overall structure. It plays a critical role in ensuring the vehicle's strength, safety, and stability. When designing a chassis frame, engineers calculate two key aspects:

Maximum Stress: The highest amount of force or pressure the frame can handle when it is loaded. How much the frame bends or deforms under load in a stationary (static) condition.

A. Existing Chassis

To perform calculations involving material properties, you need to define the material type and its specific grade, as each grade has distinct properties. The mechanical properties are essential for accurate calculations in structural analysis, stress, strain, and deformation modeling.

The length and the width of chassis frame are given in following Table.1:

No.	Description	Dimensions (mm)
1	Length	2799.68
2	Width	610

TABLE 1. Dimensions

The chassis of a vehicle supports the weight of various components, such as the driver, passengers, seats, luggage, batteries, Chassis and body. The weight of each of these elements is provided in the following Table.2, which will help in understanding the overall load distribution on the chassis frame.

No.	Component	Weight (kg)
01	Owner	68
02	Battery	70
03	People	204
04	Seats	18
05	Luggage	40
06	Frame	150

TABLE 2: Weight

Mechanical properties of different materials are given in following Table.3:

Properties	Aluminum	Magnesium Alloy	Low Alloy Steel	Grey Cast Iron	Carbon Steel
Grade	6061-T6	AZ31B	4140	ASTM A48	1020
Modulus of elasticity (E)	68.9 GPa	45 GPa	205 GPa	70-150 GPa	205 GPa
Poisson's ratio (μ)	0.33	0.35	0.27-0.30	0.25-0.30	0.29
Density(ρ)	2.70 g/cm ³	1.77 g/cm ³	7.85 g/cm ³	7.15 g/cm ³	7.87 g/cm ³
Yield strength	276 MPa	150 MPa	415 MPa	Not defined	295 MPa
Ultimate strength	310 MPa	250 MPa	655 MPa	276 MPa	395 MPa

TABLE 3: Mechanical properties of material.

B. Calculations:

The chassis is modeled as a simply supported beam with a uniformly distributed load. The total mass acting on the chassis in a static condition is 550 kg. Considering a factor of safety of 1.5, the effective total mass becomes 825 kg, which corresponds to a total load of 8093.25 N (calculated as $825 \text{ kg} \times 9.81 \text{ m/s}^2$). This total load is distributed equally across the two long members of the chassis, meaning each member bears a load of 4046.625 N. Given that the length of each beam is 2799.68 mm, the uniformly distributed load (UDL) on each member is 1.44 N/mm.

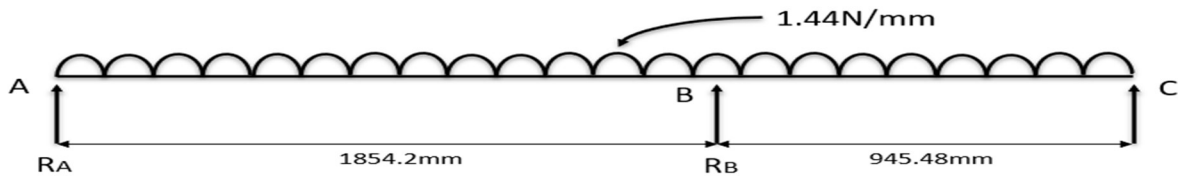


FIGURE 1. Shows that simply supported beam carried uniform distribution load of the existing chassis

C. 3D Diagram of Existing Chassis Frame:

The 3D diagram of existing chassis frame is created using CATIA software.

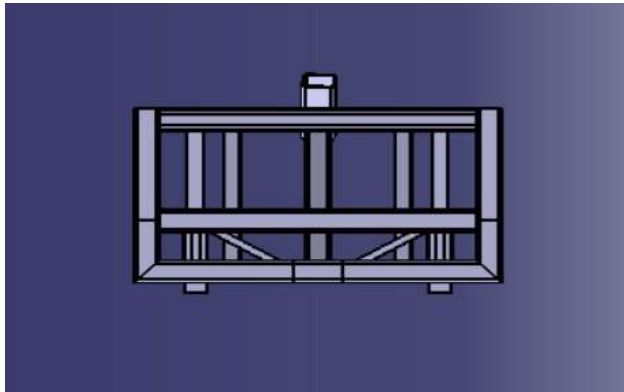


FIGURE 2(a). Front view

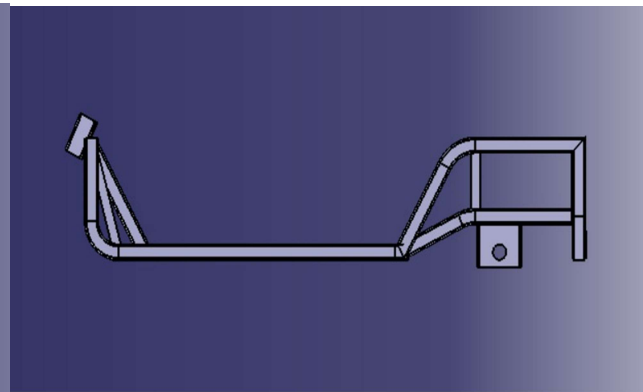


FIGURE 2(b). Side view

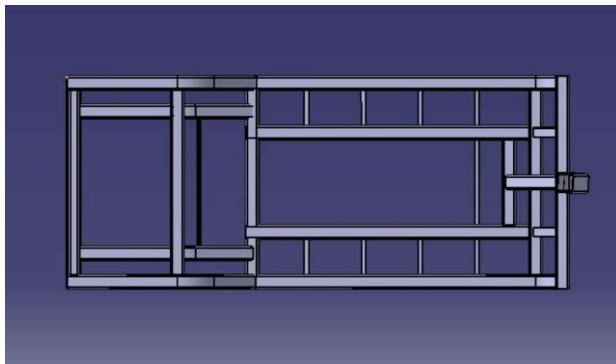


FIGURE 2(c). Top view

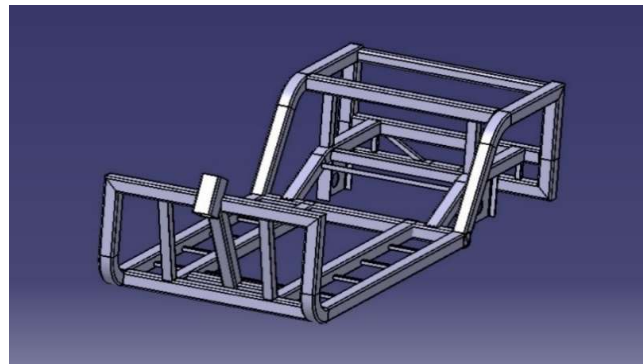


FIGURE 2(d). Assembly view of the existing chassis

D. Analysis of Existing Chassis Frame:

i) Aluminum 6061-T6

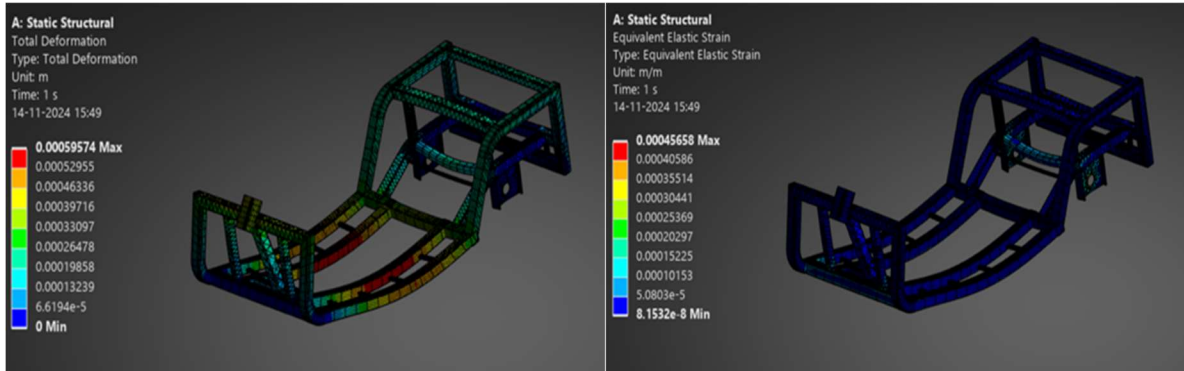


FIGURE 3(a). Total deformation

FIGURE 3(b). Strain

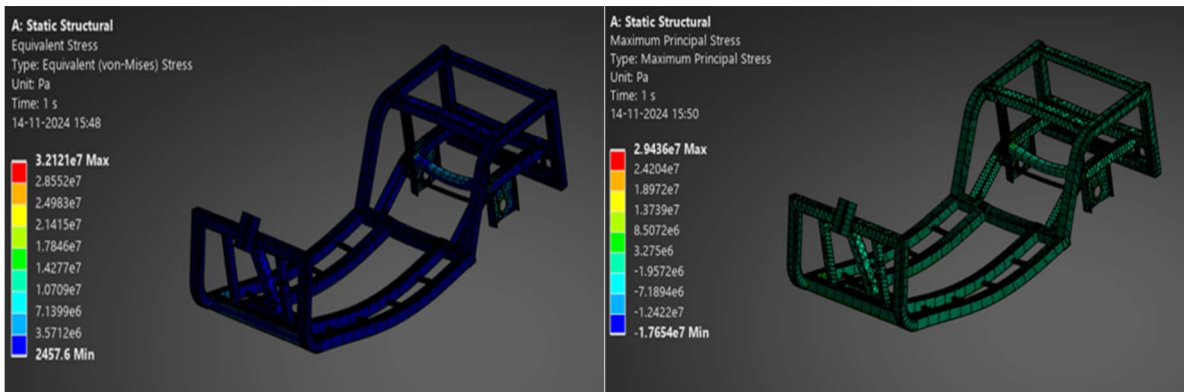


FIGURE 3(c). Stress

FIGURE 3(d). Principal stress

ii) Magnesium Alloy AZ31B

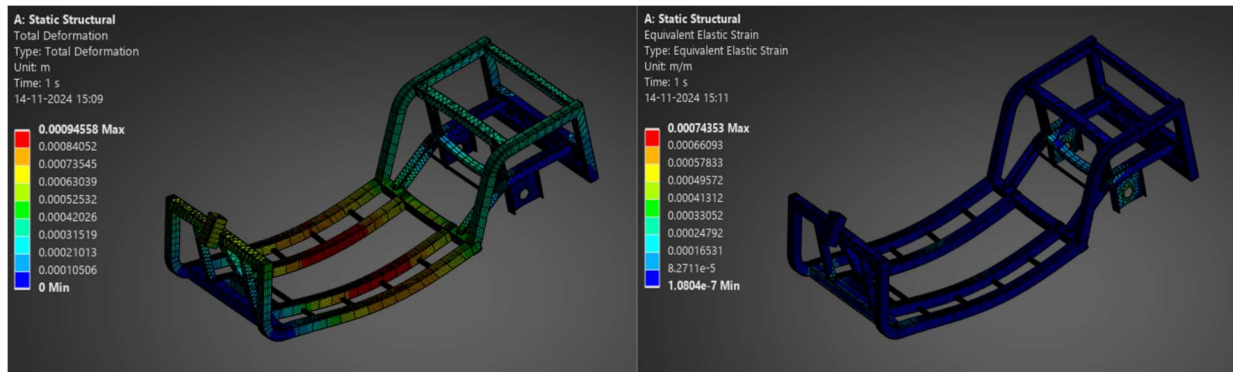


FIGURE 4(a). Total Deformation

FIGURE 4(b). Strain

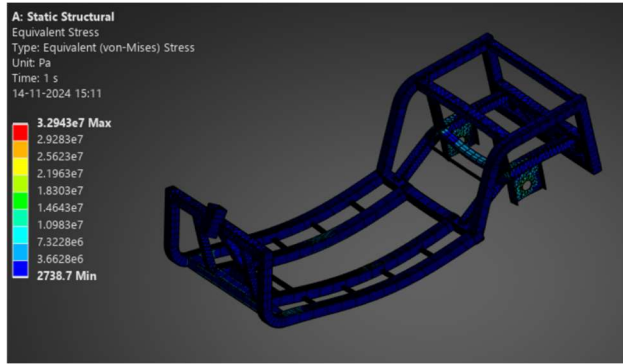


FIGURE 4(c). Stress of Magnesium Alloy Az31B

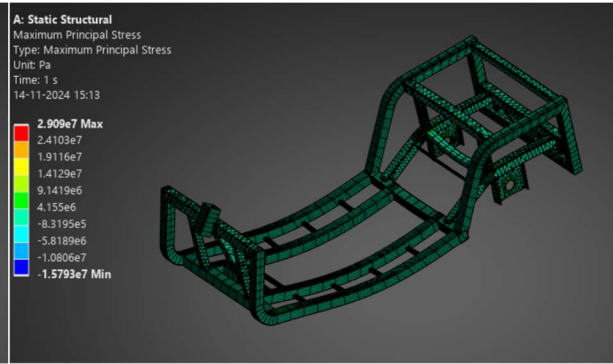


FIGURE 4(d). Principal stress of Magnesium Alloy Az31B

iii) Low Alloy Steel 4140

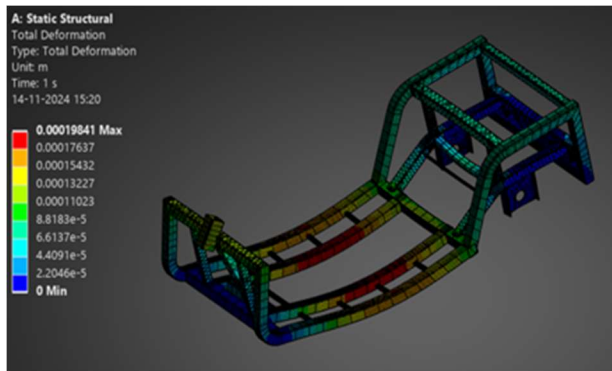


FIGURE 5(a). Total Deformation

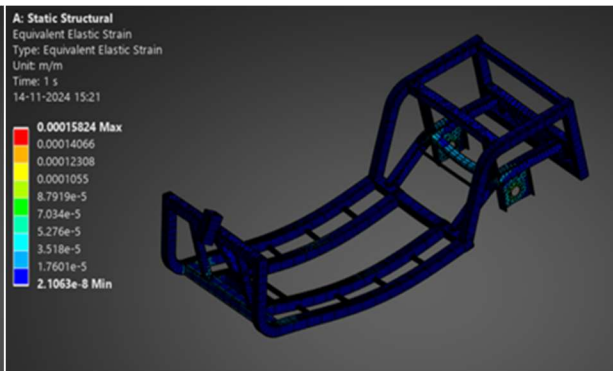


FIGURE 5(b). Strain

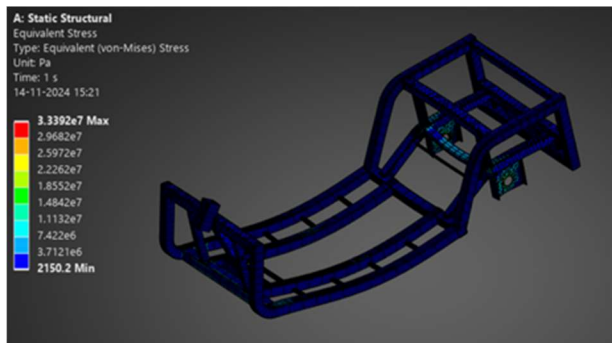


FIGURE 5(c). Stress

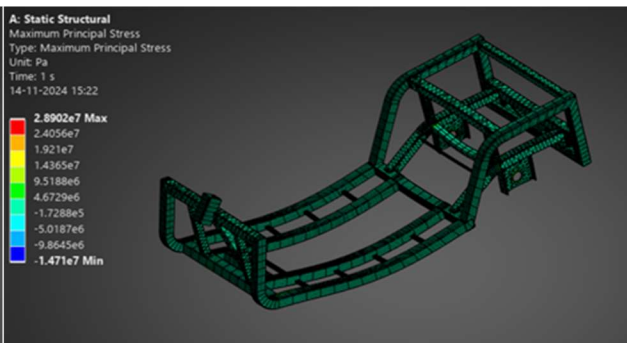


FIGURE 5(d). Principal stress

iv) Grey Cast Iron ASTM A48

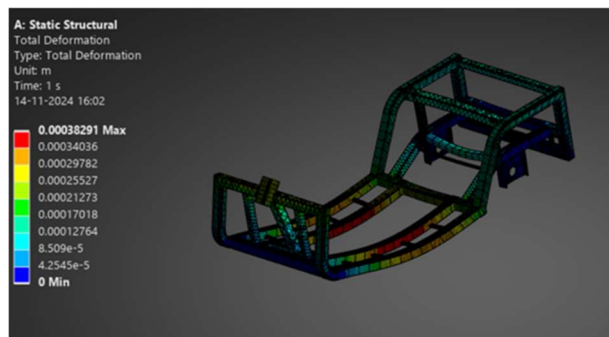


FIGURE 6(a). Total Deformation

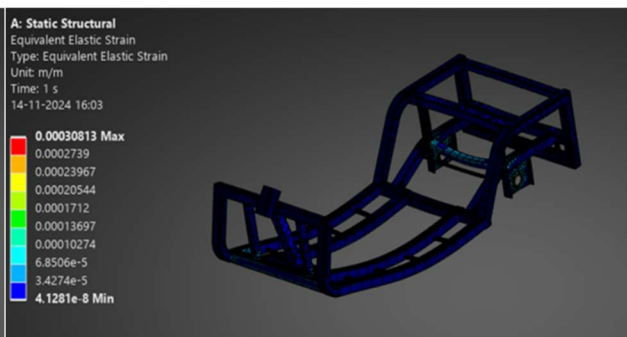


FIGURE 6(b). Strain

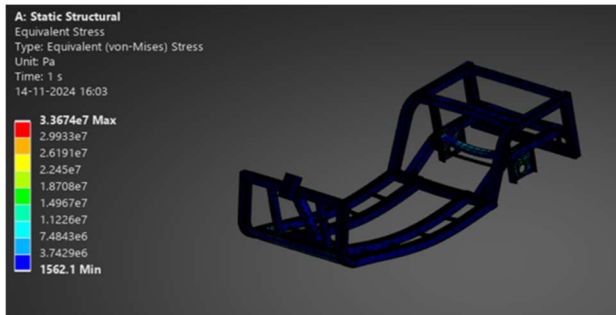


FIGURE 6(c). Stress

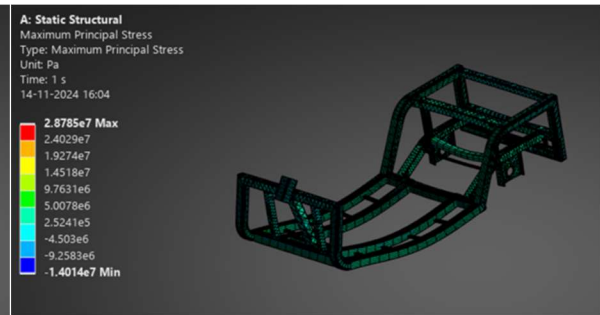


FIGURE 6(d). Principal stress

v) Carbon Steel 1020

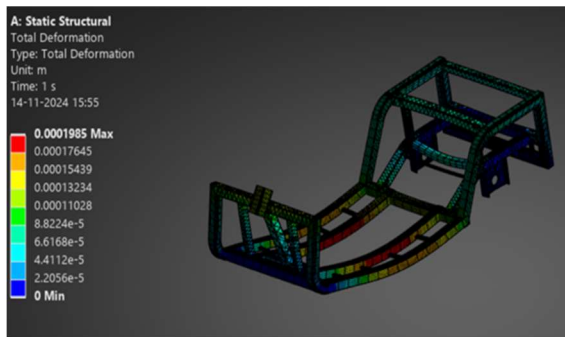


FIGURE 7(a). Total Deformation

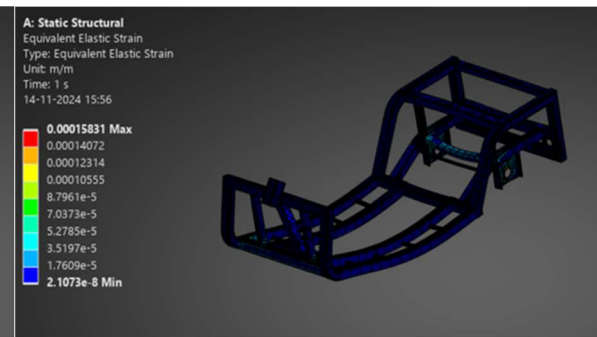


FIGURE 7(b). Strain

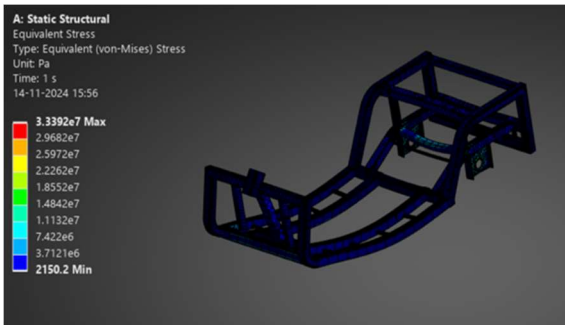


FIGURE 7(c). Stress

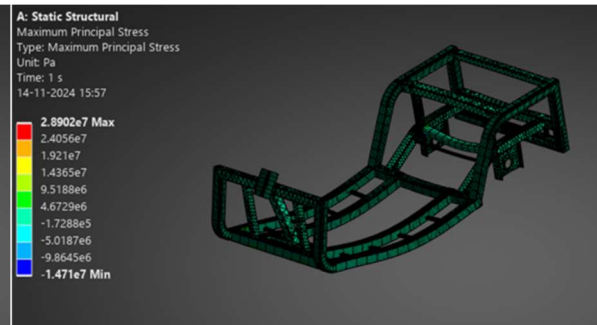


FIGURE 7(d). Principal stress

3. THEROTICAL CALCULATION OF OPTIMIZED CHASSIS:

A. Optimized Chassis:

To perform calculations involving material properties, you need to define the material type and its specific grade, as each grade has distinct properties. The mechanical properties are essential for accurate calculations in structural analysis, stress, strain, and deformation modeling.

The length and the width of chassis frame are given in following Table.4:

No.	Description	Dimensions (mm)
1	Length	2350.62
2	Width	1085

TABLE 4. Dimensions

The chassis of a vehicle supports the weight of various components, such as the driver, passengers, seats, luggage, batteries, Chassis and body. The weight of each of these elements is provided in the following Table.5, which will help in understanding the overall load distribution on the chassis frame

NO.	Component	Weight (kg)
01	Owner	68
02	Battery	70
03	People	204
04	Seats	18
05	Luggage	40
06	Frame	150

TABLE 5. Weight

Mechanical properties of different materials are given in following Table.6:

Properties	Aluminum	Magnesium Alloy	Low Alloy Steel	Grey Cast Iron	Carbon Steel
Grade	6061-T6	AZ31B	4140	ASTM A48	1020
Modulus of elasticity (E)	68.9 GPa	45 GPa	205 GPa	70-150 GPa	205 GPa
Poisson's ratio (μ)	0.33	0.35	0.27-0.30	0.25-0.30	0.29
Density(ρ)	2.70 g/cm ³	1.77 g/cm ³	7.85 g/cm ³	7.15 g/cm ³	7.87 g/cm ³
Yield strength	276 MPa	150 MPa	415 MPa	Not defined	295 MPa
Ultimate strength	310 MPa	250 MPa	655 MPa	276 MPa	395 MPa

TABLE 6. Mechanical properties of material.

B. Calculation

The chassis is modeled as a simply supported beam with a uniformly distributed load as shown Fig.28. The total mass acting on the chassis in a static condition is 550 kg. Considering a factor of safety of 1.5, the effective total mass becomes 825 kg, which corresponds to a total load of 8093.25 N (calculated as $825 \text{ kg} \times 9.81 \text{ m/s}^2$). This total load is distributed equally across the two long members of the chassis, meaning each member bears a load of 4046.625 N. Given that the length of each beam is 2350.62 mm, the uniformly distributed load (UDL) on each member is 1.72 N/mm.

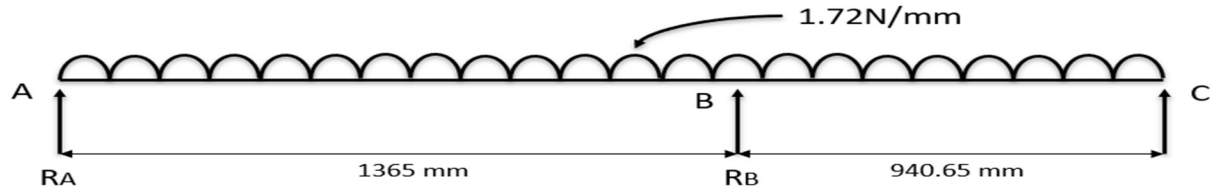


FIGURE 8. shows that simply supported beam carried uniform distribution load of the optimized chassis:

C. 3D Diagram of Optimized Chassis Frame:

The 3D diagram of optimized chassis frame is created using CATIA software.

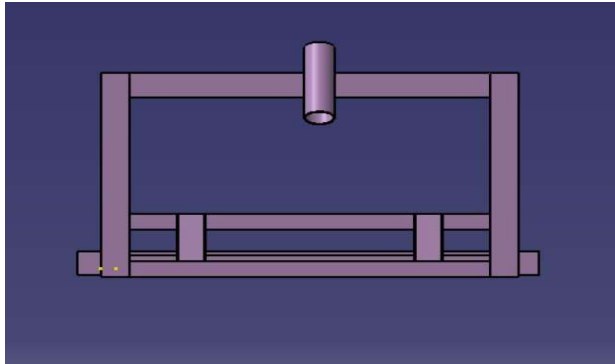


FIGURE 9(a). Front view

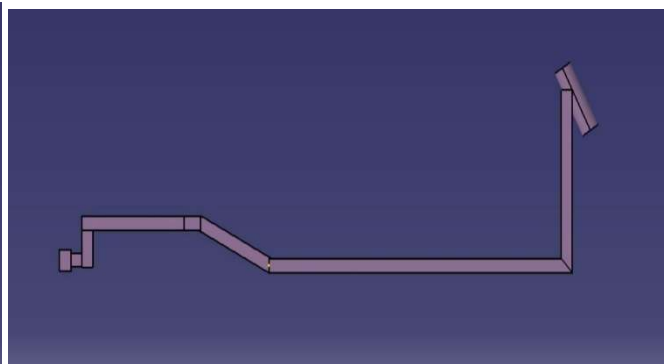


FIGURE 9(b). Side view

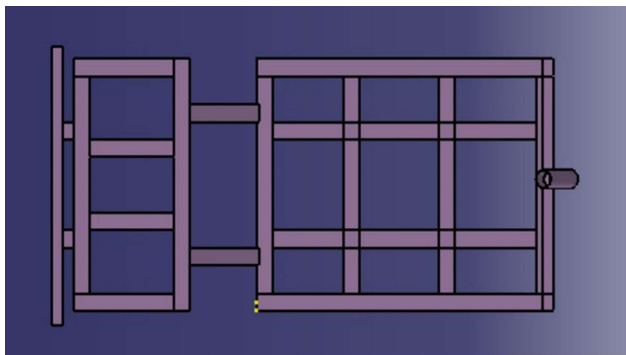


FIGURE 9(c). Top view

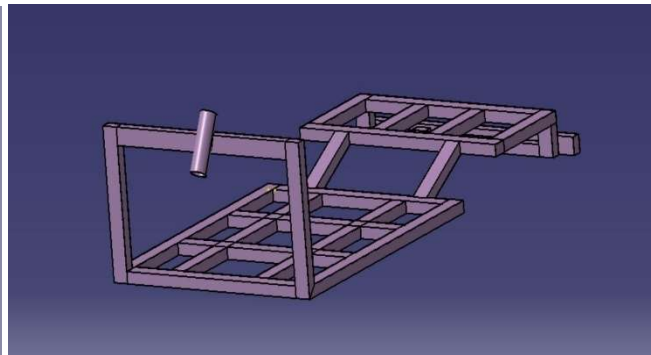


FIGURE 9(d). Assembly view

D. Analysis of Optimized Chassis Frame:

i) Alluminum 6061-T6

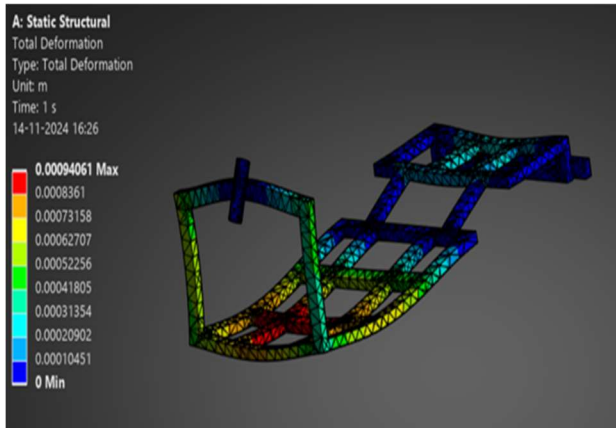


FIGURE 10(a). Total Deformation

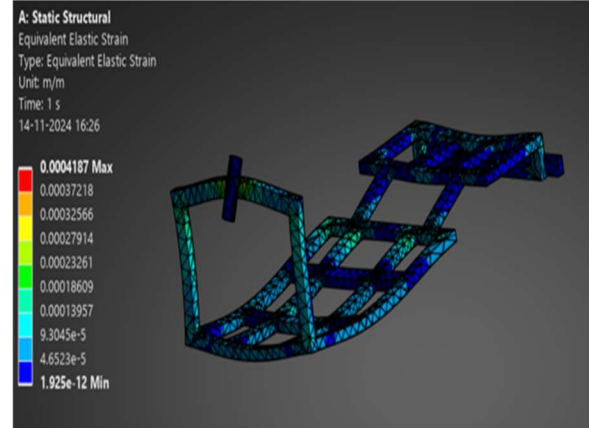


FIGURE 10(b). Strain

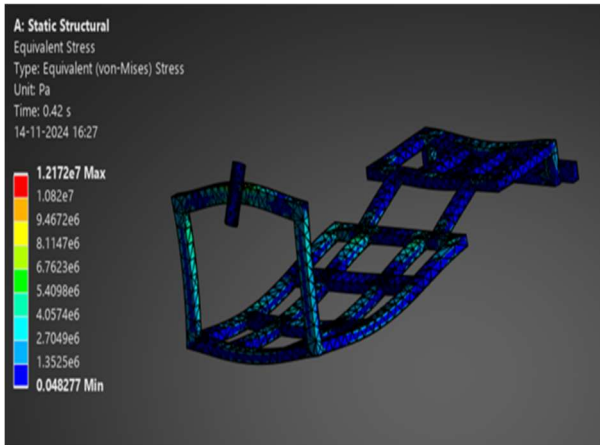


FIGURE 10(c). Stress

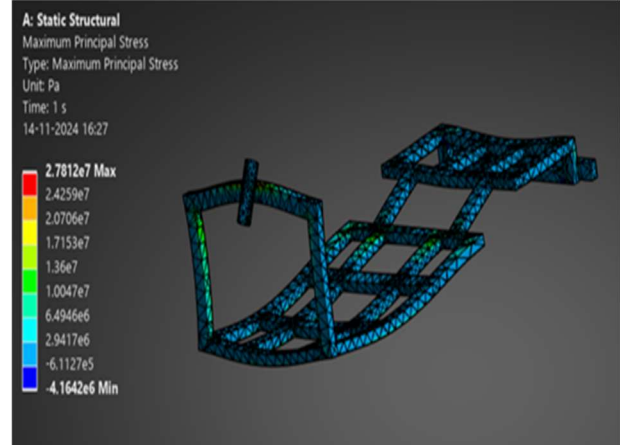


FIGURE 10(d). Principal stress

ii) Magnesium Alloy AZ31B

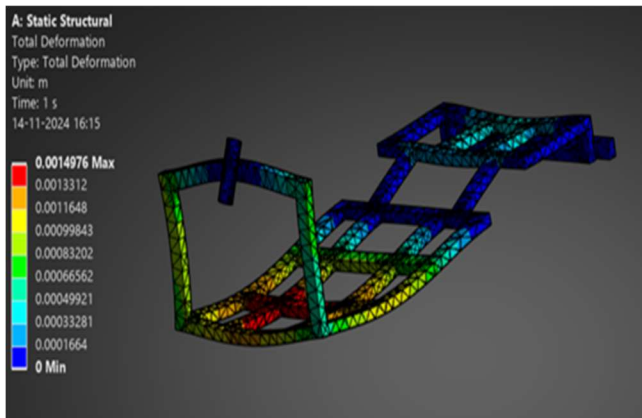


FIGURE 11(a). Total Deformation

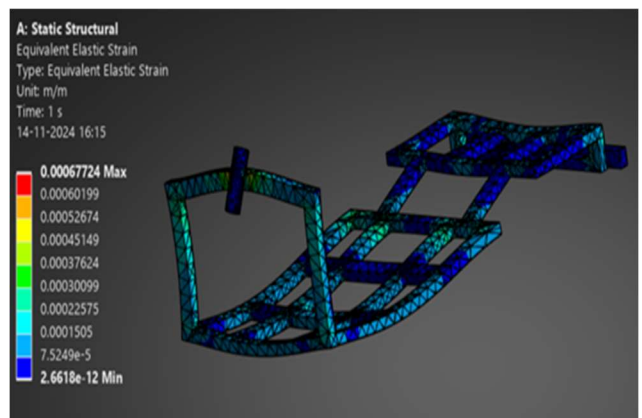


FIGURE 11(b). Strain

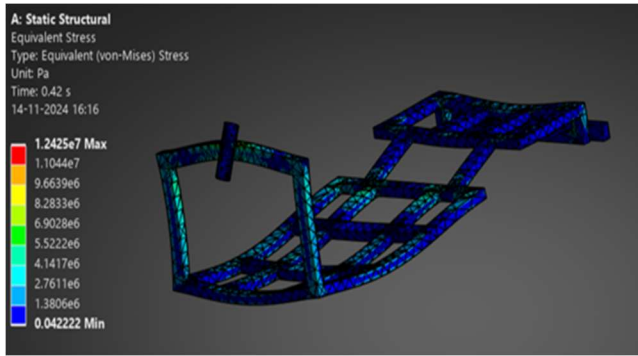


FIGURE 11(c). Stress

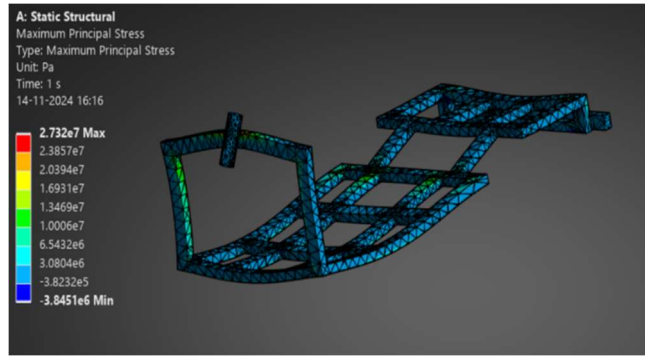


FIGURE 11(d). Principal stress

iii) Low Alloy Steel 4140

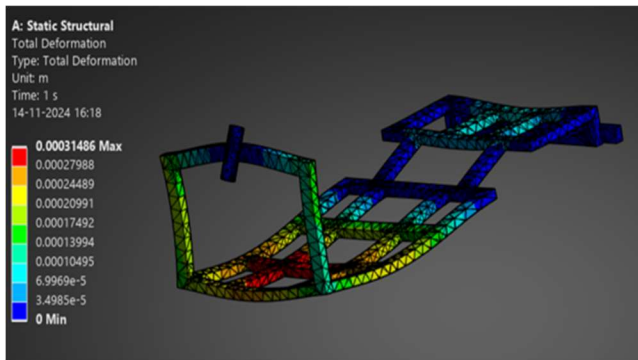


FIGURE 12(a). Total Deformation

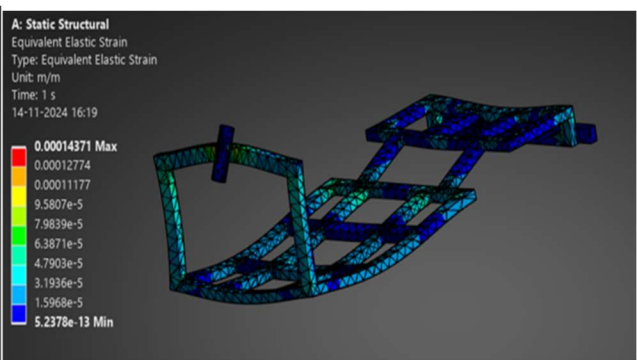


FIGURE 12(b). Strain

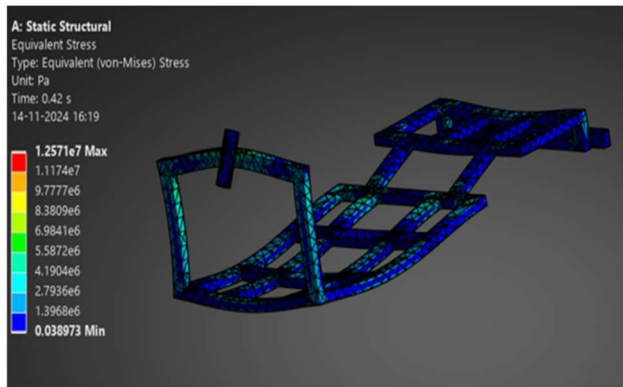


FIGURE 12(c). Stress

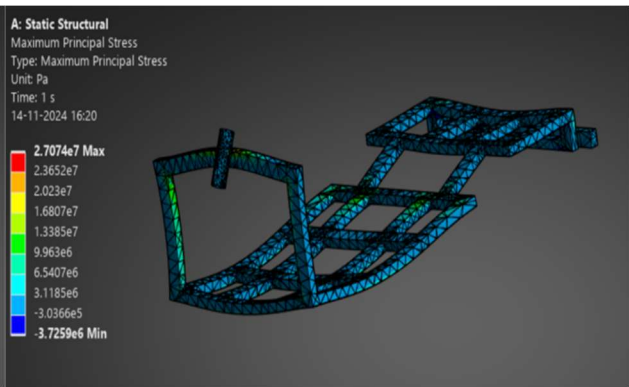


FIGURE 12(d). Principal stress

iv) Grey Cast Iron ASTM A48

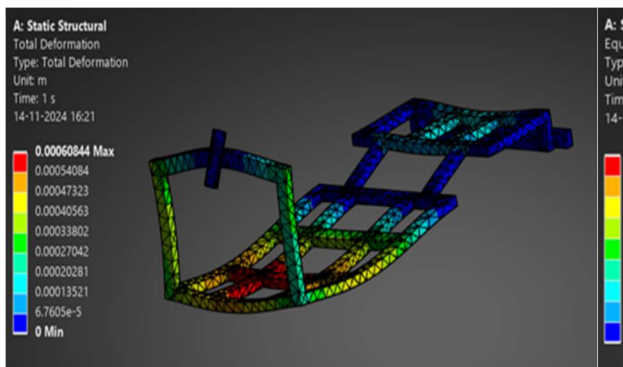


FIGURE 13(a). Total Deformation

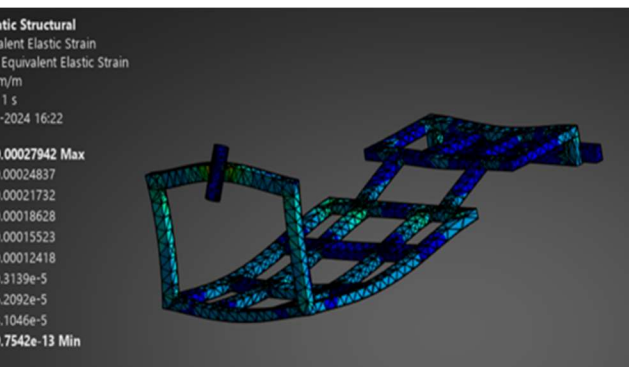


FIGURE 13(b). Strain

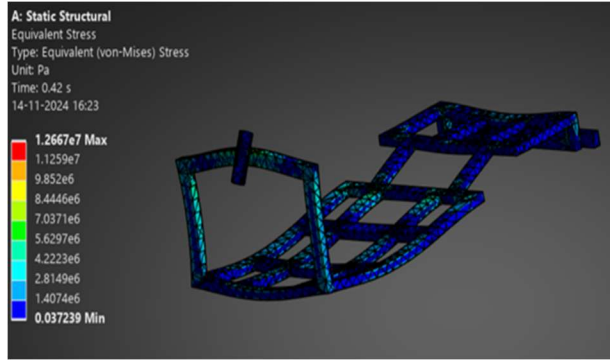


FIGURE 13(c). Stress

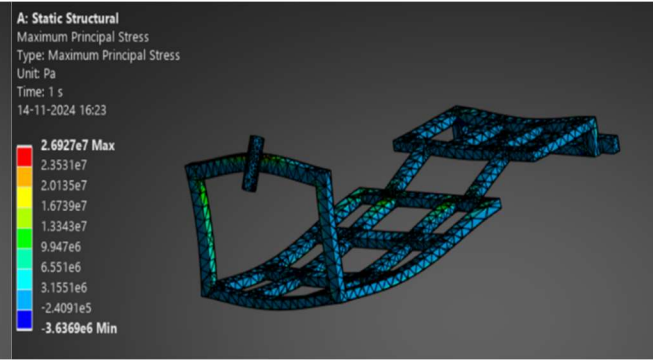


FIGURE 13(d). Principal stress

v) Carbon Steel 1020

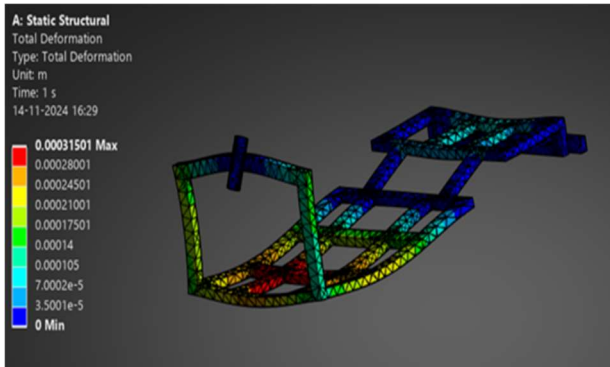


FIGURE 14(a). Total Deformation

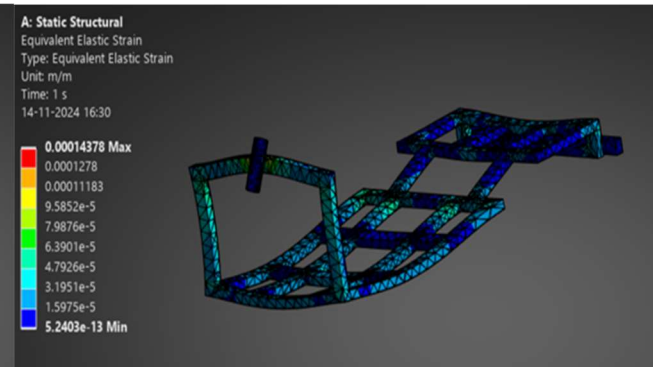


FIGURE 14(b). Strain

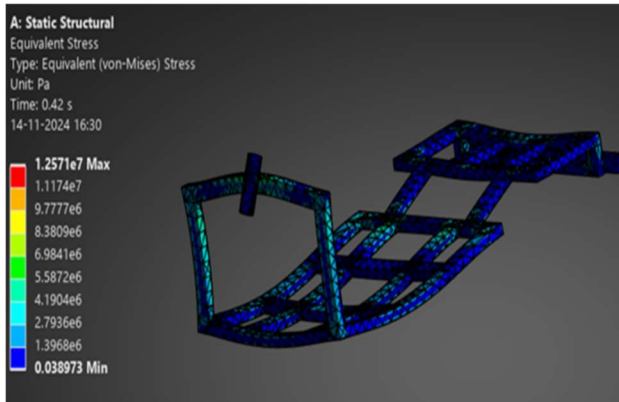


FIGURE 14(c). Stress

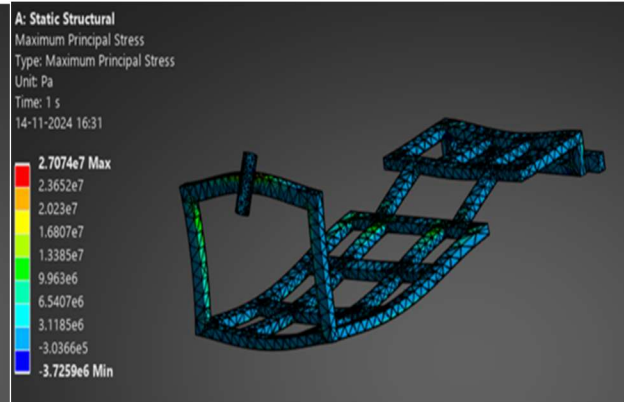


FIGURE 14(d). Principal stress

4. RESULTS

A. Existing Chassis:

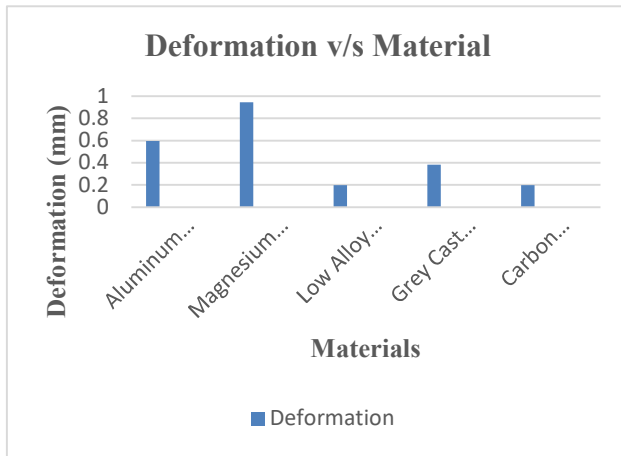


FIGURE 15(a). Deformation vs Materials

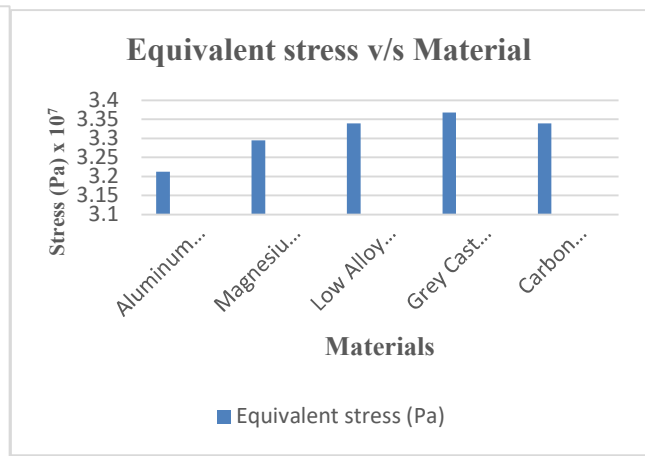


FIGURE 15(b). Stress vs Materials

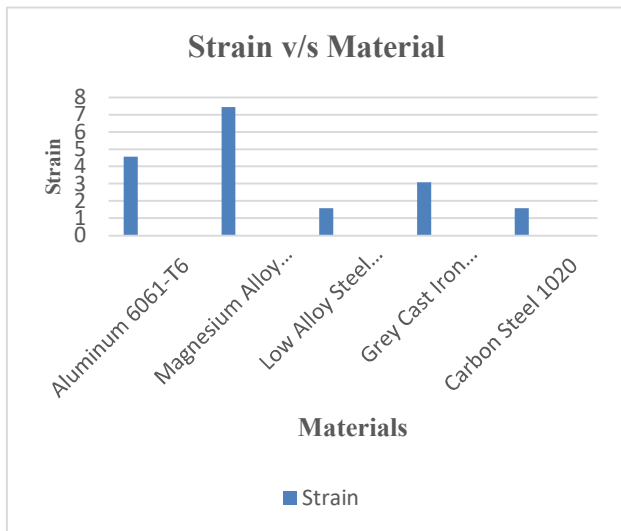


FIGURE 15(c). Strain vs Materials

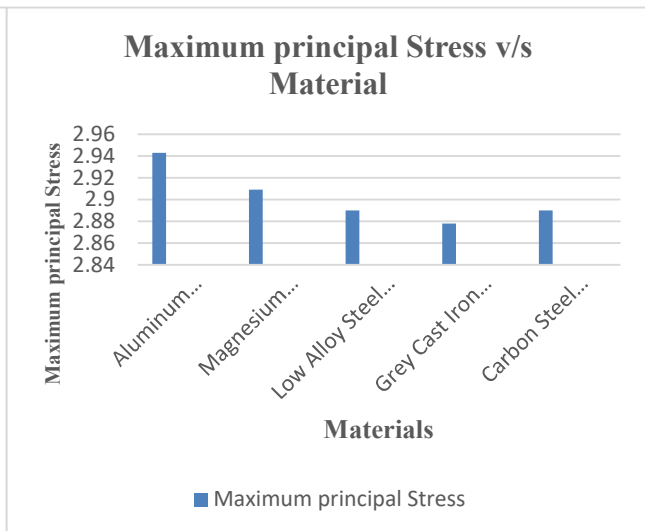


FIGURE 15(d). Principal Stress vs Materials

The Fig.15(a) illustrates the deformation values for Aluminum 6061-T6, Magnesium Alloy AZ31B, Low Alloy Steel 4140, Grey Cast Iron ASTM A48, and Carbon Steel 1020. As shown, Magnesium Alloy AZ31B experiences the highest deformation, while Low Alloy Steel 4140 exhibits the least. The Fig.15(b) illustrates the stress values for Aluminum 6061-T6, Magnesium Alloy AZ31B, Low Alloy Steel 4140, Grey Cast Iron ASTM A48, and Carbon Steel 1020. As shown, Grey Cast Iron ASTM A48 experiences the highest stress, while Aluminum 6061-T6 exhibits the least. The Fig.15(c) illustrates the strain values for Aluminum 6061-T6, Magnesium Alloy AZ31B, Low Alloy Steel 4140, Grey Cast Iron ASTM A48, and Carbon Steel 1020. As shown, Magnesium Alloy AZ31B experiences the highest strain, while Carbon Steel 1020 exhibits the least. The Fig.15(d) illustrates the principal stress values for Aluminum 6061-T6, Magnesium Alloy AZ31B, Low Alloy Steel 4140, Grey Cast Iron ASTM A48, and Carbon Steel 1020. As shown, Aluminum 6061-T6 experiences the highest principal stress, while Grey Cast Iron ASTM A48 exhibits the least.

The table 7 shows the static structural properties of various materials analyzed for their suitability in the existing chassis design, including deformation, stress, and strain:

Sl. No.	Material	Deformation (mm)	Equivalent stress (Pa)	Maximum principal Stress (Pa)	Strain
1.	Aluminum 6061-T6	0.5957	3.2121 x 10 ⁷	2.943 x 10 ⁷	4.565 x 10 ⁻⁴
2.	Magnesium Alloy AZ31B	0.9455	3.2943 x 10 ⁷	2.909 x 10 ⁷	7.435 x 10 ⁻⁴
3.	Low Alloy Steel 4140	0.1984	3.3392 x 10 ⁷	2.890 x 10 ⁷	1.584 x 10 ⁻⁴
4.	Grey Cast Iron ASTM A48	0.3829	3.3674 x 10 ⁷	2.878 x 10 ⁷	3.081 x 10 ⁻⁴
5.	Carbon Steel 1020	0.1985	3.3392 x 10 ⁷	2.890 x 10 ⁷	1.583 x 10 ⁻⁴

TABLE 7. Static Structural Results for existing chassis

B. Optimized Chassis:

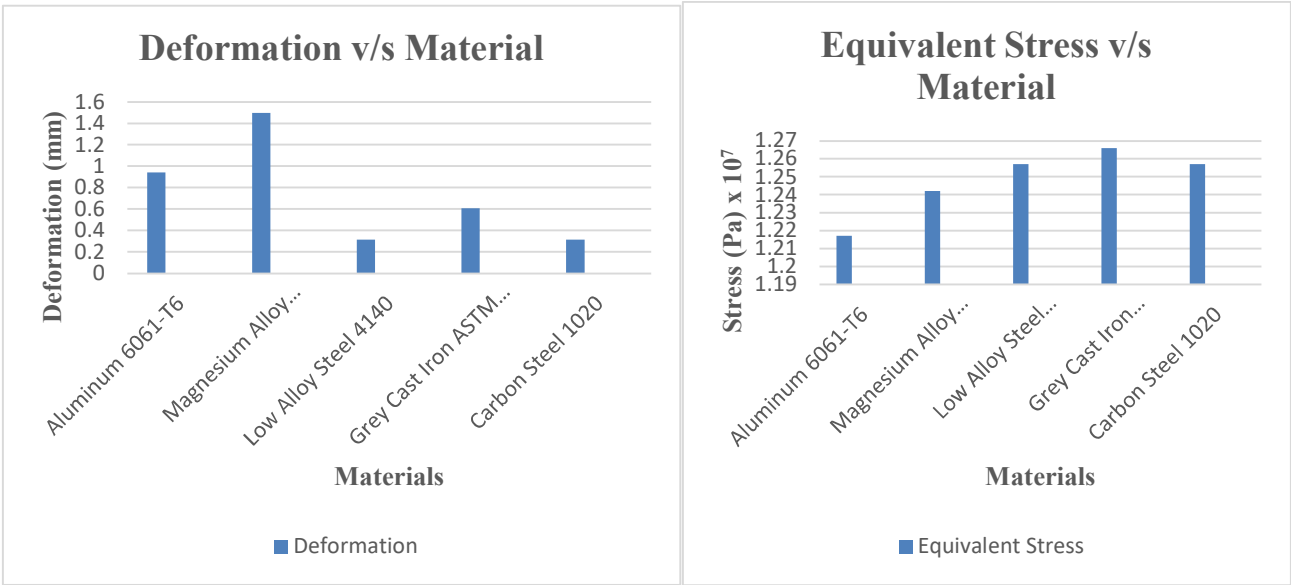


FIGURE 16(a). Deformation vs Materials

FIGURE 16(b). Stress vs Materials

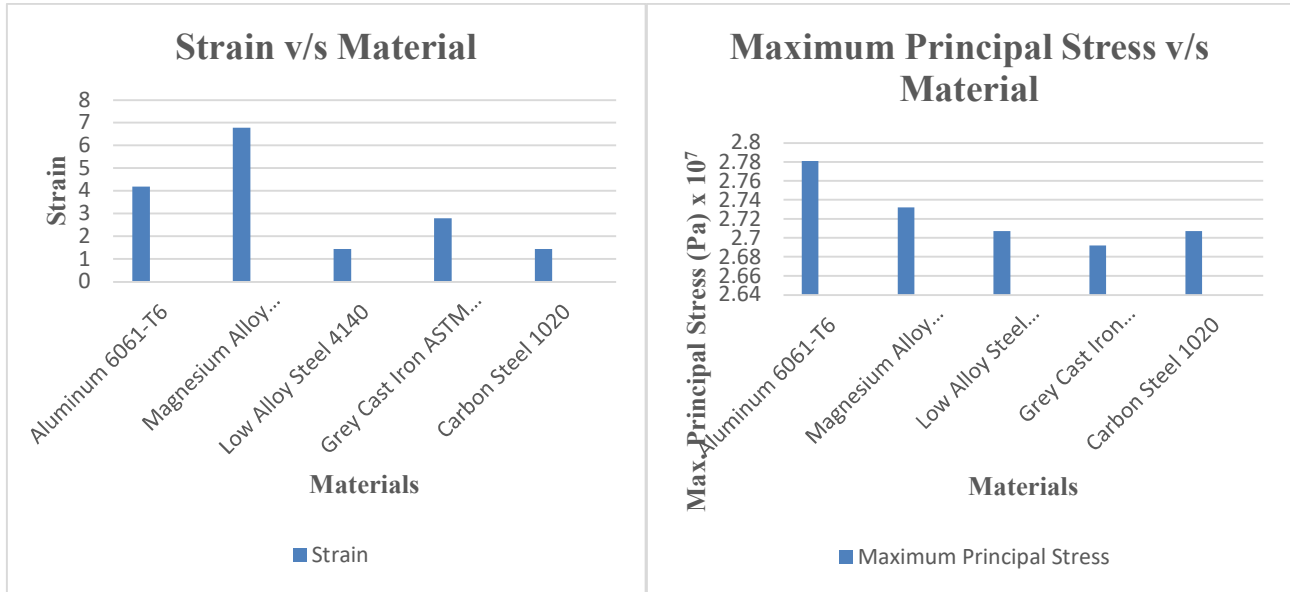


FIGURE 15(c). Strain vs Materials

FIGURE 15(d). Principal Stress vs Materials

The Fig.16(a) illustrates the deformation values for Aluminum 6061-T6, Magnesium Alloy AZ31B, Low Alloy Steel 4140, Grey Cast Iron ASTM A48, and Carbon Steel 1020. As shown, Magnesium Alloy AZ31B experiences the highest deformation, while Low Alloy Steel 4140 exhibits the least. The Fig.16(b) illustrates the stress values for Aluminum 6061-T6, Magnesium Alloy AZ31B, Low Alloy Steel 4140, Grey Cast Iron ASTM A48, and Carbon Steel 1020. As shown, Grey Cast Iron ASTM A48 experiences the highest stress, while Aluminum 6061-T6 exhibits the least. The Fig.16(c) illustrates the strain values for Aluminum 6061-T6, Magnesium Alloy AZ31B, Low Alloy Steel 4140, Grey Cast Iron ASTM A48, and Carbon Steel 1020. As shown, Magnesium Alloy AZ31B experiences the highest strain, while Carbon Steel 1020 and Low Alloy Steel 4140 exhibits the least. The Fig.16(d) illustrates the principal stress values for Aluminum 6061-T6, Magnesium Alloy AZ31B, Low Alloy Steel 4140, Grey Cast Iron ASTM A48, and Carbon Steel 1020. As shown, Aluminum 6061-T6 experiences the highest principal stress, while Grey Cast Iron ASTM A48 exhibits the least.

The table 8 shows the static structural properties of various materials analyzed for their suitability in the optimized chassis design, including deformation, stress, and strain:

Sl. No.	Material	Deformation (mm)	Equivalent stress (Pa) 10 ⁷	Maximum principal Stress (Pa)	Strain
1.	Aluminum 6061-T6	0.940	1.217 x 10 ⁷	2.781 x 10 ⁷	4.187 x 10 ⁻⁴
2.	Magnesium Alloy AZ31B	1.497	1.242 x 10 ⁷	2.732 x 10 ⁷	6.772 x 10 ⁻⁴
3.	Low Alloy Steel 4140	0.314	1.257 x 10 ⁷	2.707 x 10 ⁷	1.437 x 10 ⁻⁴
4.	Grey Cast Iron ASTM A48	0.608	1.266 x 10 ⁷	2.692 x 10 ⁷	2.794 x 10 ⁻⁴
5.	Carbon Steel 1020	0.315	1.257 x 10 ⁷	2.707 x 10 ⁷	1.437 x 10 ⁻⁴

TABLE 8. Static Structural Results for Optimized chassis

The table 9 compares the structural and dimensional characteristics of the existing chassis with those of the optimized chassis, highlighting improvements and differences in performance parameters:

Existing Chassis	Optimized Chassis
1.Overall length=2799.68mm	1.Overall length=2350.62mm
2.Overall width=610mm	2.Overall width=1085mm
3.Uniform distributed load=1.44 N/mm	3.Uniform distributed load=1.72 N/mm
4.Average deformation=0.464mm	4.Average deformation=0.7353mm
5. Average stress=33.718 N/mm ²	5. Average stress=12.481 N/mm ²
6. Average strain=0.3646	6. Average strain=0.3326
7. Average principal stress=29.023 N/mm ²	7. Average principal stress=27.241 N/mm ²

TABLE 9. Existing Chassis vs Optimized Chassis

The table 10 shows the Comparison between different materials used in Optimized chassis on the bases of deformation, stress and strain:

Material	Deformation	Strain	Stress	Principal Stress
Aluminum 6061-T6	0.00094061m	0.0004187	1.2172*10 ⁷ N/mm ²	2.7812*10 ⁷ N/mm ²
Magnesium Alloy AZ31B	0.0014976m	0.00067724	1.2425*10 ⁷ N/mm ²	2.732*10 ⁷ N/mm ²
Low Alloy Steel 4140	0.00031486m	0.00014371	1.2571*10 ⁷ N/mm ²	2.7074*10 ⁷ N/mm ²
Gray Cast Iron ASTM A48	0.00060844m	0.00027942	1.2667*10 ⁷ N/mm ²	2.6927*10 ⁷ N/mm ²
Carbon Steel 1020	0.00031501m	0.00014378	1.2571 *10 ⁷ N/mm ²	2.7074*10 ⁷ N/mm ²

TABLE 10. Different materials used in Optimized chassis

5. CONCLUSION

The optimized chassis is better than the existing chassis in several ways. It can handle more load (1.72 N/mm vs. 1.44 N/mm), experiences less stress (12.481 N/mm² vs. 33.718 N/mm²), and has slightly less strain (0.3326 vs. 0.3646), showing it is stronger and more efficient. Its shorter length and wider design improve stability and handling. However, it deforms more (0.7353 mm vs. 0.464 mm), which makes it more flexible and good for absorbing shocks. Overall, the new chassis is a better design for modern needs. Among the materials analyzed, Low Alloy Steel 4140 and Carbon Steel 1020 exhibit the least deformation and strain, making them the most rigid and suitable for applications requiring high stiffness. Conversely, Magnesium Alloy AZ31B shows the highest deformation and strain, indicating lower rigidity but potential advantages in weight-sensitive applications. Aluminum 6061-T6 experiences the highest principal stress, which may limit its performance under stress concentrations, while Gray Cast Iron ASTM A48 demonstrates good stress resistance but could be unsuitable for applications needing ductility due to its brittleness. Overall, material selection depends on specific application needs, with steel alloys favored for rigidity and durability and lightweight alloys like aluminum and magnesium for applications prioritizing weight reduction. Low Alloy Steel 4140 is the best material overall due to its combination of low deformation, low strain, and high rigidity, making it ideal for applications requiring strength, durability, and resistance to deformation under load. It provides excellent performance in demanding conditions compared to lighter but less rigid materials like aluminum, magnesium, gray cast iron or carbon steel.

6. REFERENCES

1. Akshay Sutar Dr. Sanjay D. Yadav, Static Analysis of Chassis Frame of Battery-Operated Three-Wheeler, IJSRD - International Journal for Scientific Research & Development| Vol. 8, Issue 8, 2020 | ISSN (online): 2321-0613.
2. Miroslav Blatnický, Ján Dižo, Andrej Suchánek and Pavol Šťastniak. Research on dynamical properties of a three wheeled electric vehicle from the point of view of driving safety. Open Engineering 2023; 13; 20220441
3. Gülüstan Tuğçe Alvalı, Ali Balbay, Serkan Güneş, Burak Yenipınar, M. Cem Çatalbaş, Turan Şişman. Design and Implementation of a Three-Wheel Multi-Purpose Electric Vehicle with Finite Elements Analysis, International Journal of Automotive Science and Technology, May 2021.
4. Tamer Nabil, Basem E. El-Naghi, Mahmoud Saeed, Ahmed Kamal, Eslam Gharib, Mohanad Mohsen, and Ibrahim Ahmed, Design and Fabrication of Prototype Battery Electric Three Wheeled Vehicles, Journal of Asian Electric Vehicles, Volume 17, Number 2, December 2019.
5. Ioan Szabo , Liviu . Scurtu , Horia Raboca and Florin Mariasiu, Topographical Optimization of a Battery Module Case That Equips an Electric Vehicle, International Society for Porous Media, Batteries 2023, 9, 77.
6. Shashank.N , Shashank.P , Gopalakrishna, Design and Analysis of Skateboard Electric Vehicle chassis and Battery Pack, International Research and technology, Volume 12, issue 8 August 2023.
7. Bharati Parmar, Sachin Rajput, Pankaj Sanap, Dr. Fauzia Siddiqui, Design of a Three-Wheeler Electric Vehicle, International Journal of Scientific and Engineering Research, Volume 9, Issue 5 May 2018.
8. Suvac, L Cucu, M Stoica, M Savaniu , Development and optimization of a three-wheeled electric vehicle frame using structural materials, Journal of Physics, Volume 1, Issue 1 January 2021.
9. Mohammad Waseem, Ahmad Faizan Sherwani, and Mohd Suhaib, Driving Pattern-based Optimization and Design of Electric Propulsion System for Three-Wheeler Battery Vehicle, International Journal of Performability Engineering, Volume 16, Number 3, Issue March 2020.
10. Yash Parihar, OM Prakash Shukla and Jaydeep Shah, Optimization and analysis of chassis for heavy electric truck under different condition, World Journal of Advanced Engineering Technology and Science, Volume 3, Issue 18 May 2024.
11. Mohammed Noorul Hussain, Mohammed Zahed, Mohammed Omer Farooq, Design, Analysis and Fabrication of 3-Wheeled Hybrid Vehicle Run by Human Effort and Electric Motor, International Journal of Engineering Research and Technology, Volume 3, Issue 1 January 2014.
12. Chien-Cheng Lin, Song-Jeng Huang and Chi-Chia Liu, Structural analysis and optimization of bicycle frame designs, Advances in Mechanical Engineering 2017, Volume 9(12), Issue 5 October 2017.
13. S.NAVEEN KUMARI ,R.KARTHIK,S.KANISHKAR ,M.VIGNESWARAN, S.SYEDFAKHRUDEEN, Design and Analysis of Automobile Chassis by using Alternative Materials, Journal of Xi'an Shiyu University, Natural Science Edition, ISSN: 1673-064X.
14. Dr. Pranav Charkha, Dr.Nitin Sherje, Mr. Santosh Dabhole, Mr. Rakesh Raushan, Mr. Ashish Ekatpure, Mr. Dinesh Burande,Dr. Prasad Dhore,Mr. Jitendra Narkhede, Comparative Analysis of Materials for Chassis Design in a Three-Wheeled Electric Vehicle, Advances in Nonlinear Variational Inequalities, ISSN: 1092-910X,Vol 27 No. 3 (2024).
15. N.S. Prasanna Rao, P. Sunil Kumar, MODAL ANALYSIS OF THREE-WHEELER AUTOCHASSIS USING ANSYS, International Journal of Mechanical Engineering and Technology, Volume 10, Issue 2, February 2019.
16. Bubesh Kumar, Design and Fabrication of Electric Three-Wheeler Vehicle, Research and Developments in Engineering Research Vol. 4, issue 30 May 2023
17. R. Nilesh Singh, S.Rudraksh, M.Sachidhanandam, Design, Development and Analysis of a Three-Wheeler Vehicle Chassis, International Journal of Vehicle Structures and Systems, Vol. 14, issue 30 January 2022.
18. Sarvpreeth Vinayaka P S Vinyas G, Balaji G, H R Chandan Kumar, Sharanbassappa S. Patil, Design and Development of Chassis for Front-Wheel Drive Electric Tadpole Vehicle, International Journal of Engineering Research in Mechanical and Civil Engineering, Vol 10, Issue 5, May 2023.